

# Class 14: LU Decomposition

If you need to solve many equations of the form  $A\mathbf{x} = \mathbf{b}_1$ ,  $A\mathbf{x} = \mathbf{b}_2$ , ...,  $A\mathbf{x} = \mathbf{b}_p$  but do not have all the  $\mathbf{b}$ 's at the same time it is efficient to factor the matrix  $A$  into the product of a lower triangular matrix  $L$  and an upper triangular matrix  $U$ .

Overall strategy: Given  $A$  is an  $m \times n$  matrix

Step 1

Step 2

Example: Given  $\begin{bmatrix} 1 & -1 & 3 \\ 2 & 0 & 6 \\ -1 & 7 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 3 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$  Solve  $A\mathbf{x} = \begin{bmatrix} 8 \\ 16 \\ -2 \end{bmatrix}$



Key idea: Given the factorization  $A = LU$ , you can solve the equation  $A\mathbf{x} = \mathbf{b}$  in two quick steps.

1.

2.

Three big questions:

1.

2.

3.

Example

$$\begin{bmatrix} 1 & 3 & 4 & 0 & 0 \\ -3 & -6 & -7 & 2 & 2 \\ 3 & 3 & 0 & -1 & -3 \\ -5 & -3 & 0 & 8 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 4 & 0 & 0 \\ 0 & 3 & 5 & 2 & 2 \\ 0 & -6 & -12 & -1 & -3 \\ 0 & 12 & 20 & 8 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 4 & 0 & 0 \\ 0 & 3 & 5 & 2 & 2 \\ 0 & 0 & -2 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Practice: Use the algorithm to find an  $LU$  factorization for each matrix below.

$$\begin{bmatrix} 2 & 1 & 1 \\ 4 & 4 & 4 \\ 2 & -3 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 2 & 1 & 2 \\ 6 & 6 & 5 & 5 \\ 0 & -4 & -6 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 2 \\ 8 & 7 \\ 12 & 6 \\ 16 & 5 \end{bmatrix}$$